

Complex Analysis

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1 Complex Number System

1.1 Algebraic Structure

Definition 1.1

Complex Numbers

The set of complex numbers $\mathbb{C}$ is $\mathbb{R}^2$, together with the operations

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

and

$$(x_1, y_1) \cdot (x_2, y_2) = (x_1x_2 - y_1y_2, x_1y_2 + x_2y_1)$$

Theorem 1.1

Field Structure

$\mathbb{C}$ is a field. With $0 = (0, 0)$, $-(x, y) = (-x, -y)$, $1 = (1, 0)$ and

$$(x, y)^{-1} = \left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2} \right) \text{ for } (x, y) \neq (0, 0)$$

Notation 1.1

If $z = (x, y)$ we call x the “real part” of z and write $x = \operatorname{Re}(z)$.

We call y the “imaginary part” of z and write $y = \operatorname{Im}(z)$.

Define $i = (0, 1)$.

Remark 1.2

There is an embedding

$$\begin{aligned} \mathbb{R} &\hookrightarrow \mathbb{C} \\ x &\mapsto (x, 0) \end{aligned}$$

so we view $\mathbb{R} \subseteq \mathbb{C}$.

Notation 1.3

Under the above embedding, any $z \in \mathbb{C}$ can be written as $z = \operatorname{Re}(z) + \operatorname{Im}(z) \cdot i$ and $i^2 = -1$

1.2 Conjugation and Polar Form

Definition 1.2

Conjugation

Define *conjugation* to be the map

$$\begin{aligned} - : \mathbb{C} &\longrightarrow \mathbb{C} \\ z = x + yi &\longmapsto x - yi = \bar{z} \end{aligned}$$

Note

$$\operatorname{Re}(z) = \frac{z + \bar{z}}{2}, \quad \operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$$

Note

If $z = x + yi \in \mathbb{C}$ it has a polar representation $(x, y) \mapsto (r, \theta)$ via $z = r \cos \theta + ir \sin \theta$ where $r = \sqrt{x^2 + y^2} \geq 0$, $x = r \cos \theta$, $y = r \sin \theta$. The angle θ is defined modulo 2π when $z \neq 0$, and is arbitrary when $z = 0$.

Remark 1.4

If z_1, z_2 have polar representations

$$z_1 : (r_1, \theta_1), \quad z_2 : (r_2, \theta_2)$$

Then

$$\begin{aligned} z_1 \cdot z_2 &= (r_1 \cos \theta_1 + ir_1 \sin \theta_1) \cdot (r_2 \cos \theta_2 + ir_2 \sin \theta_2) \\ &= r_1 r_2 ((\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2)) \\ &= r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)) \end{aligned}$$

1.3 Topology and Argument**Note**

If $z = x + iy$ and $r = \sqrt{x^2 + y^2}$, then $r^2 = z\bar{z}$.

Remark 1.5

$\mathbb{C}$ is a metric space and thus the triangle inequality holds.

Remark 1.6

$\mathbb{C}$ is a topological space, so topological notions make sense.

Definition 1.3**Argument**

If $z \in \mathbb{C} \setminus \{0\}$ has polar representation (r, θ) , we call θ the *argument* of z and write $\theta = \arg(z)$. We'll think about $\arg$ as a multivalued function. We write $\operatorname{Arg}(z) = \theta$ if $\theta \in (-\pi, \pi]$, the *principal value*.

Example 1.1

If $z = r(\cos \theta + i \sin \theta)$ and $z \neq 0$ then we can find an n^{th} root of z (actually n of them).

2 Differentiation

2.1 Complex Derivatives

Definition 2.1

Complex Derivative

Suppose $f : U \rightarrow \mathbb{C}$ where $U \subseteq \mathbb{C}$ is open and contains z_0 . We'll say that f has derivative $f'(z_0)$ at z_0 if

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

Theorem 2.1

Product Rule

If f and g are differentiable at z_0 then so is fg and

$$(f \cdot g)'(z_0) = f'(z_0)g(z_0) + f(z_0)g'(z_0)$$

2.2 Cauchy-Riemann Equations

Remark 2.1

If $f : \mathbb{C} \rightarrow \mathbb{C}$, then we may write $f(z) = u(z) + iv(z)$ where $u, v : \mathbb{C} \rightarrow \mathbb{R}$. If f is differentiable at $z_0 = x_0 + iy_0$ (say $f'(z_0) = a + ib$), then

$$\begin{aligned} R(z) &= f(z) - f(z_0) - f'(z_0)(z - z_0) \\ &= u(z) - u(z_0) - a(x - x_0) + b(y - y_0) + i(v(z) - v(z_0) - b(x - x_0) - a(y - y_0)) \\ &= R_1(z) + iR_2(z) \end{aligned}$$

Then $\lim_{z \rightarrow z_0} R_1 \frac{z}{|z - z_0|} = \lim_{z \rightarrow z_0} R_2 \frac{z}{|z - z_0|} = 0$. Thus both u and v are differentiable and $\frac{\partial u}{\partial x}(z_0) = a$, $\frac{\partial u}{\partial y}(z_0) = -b$, $\frac{\partial v}{\partial x}(z_0) = b$, $\frac{\partial v}{\partial y}(z_0) = a$.

Theorem 2.2

Cauchy-Riemann Equations

If $f : \mathbb{C} \rightarrow \mathbb{C}$, $f(z) = u(z) + iv(z)$, then f is differentiable at z_0 (as a complex function) iff both u and v are differentiable at z_0 and

$$\begin{aligned} \frac{\partial u}{\partial x}(z_0) &= \frac{\partial v}{\partial y}(z_0) \quad \text{and} \\ \frac{\partial v}{\partial x}(z_0) &= -\frac{\partial u}{\partial y}(z_0) \end{aligned}$$

These are called the Cauchy-Riemann equations.

Note

If $u : \mathbb{R}^2 \rightarrow \mathbb{R}$, then u is differentiable at (x_0, y_0) if $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ exist and are continuous in neighborhoods of (x_0, y_0) .

2.3 Holomorphic Functions and Exponential

Definition 2.2

Exponential function

The exponential function is defined to be the function

$$\exp(x + iy) = e^x(\cos y + i \sin y)$$

we write $e^z := \exp(z)$.

Remark 2.2

For $y \in \mathbb{R}$, we have $e^{iy} = \cos(y) + i \sin(y)$.

Definition 2.3

Holomorphic

If $f : U \rightarrow \mathbb{C}$ with $U \subseteq \mathbb{C}$ open, such that f is differentiable at each point in U , then f is *holomorphic*.

Definition 2.4

Entire

$f : \mathbb{C} \rightarrow \mathbb{C}$ is called *entire* if it is holomorphic on all of $\mathbb{C}$.

2.4 Logarithms and Branches

Definition 2.5

Logarithm

A complex number w is a *logarithm* of z if $e^w = z$.

Remark 2.3

The range of the exponential is $\mathbb{C} \setminus \{0\}$. So there is no logarithm of 0.

Definition 2.6

Branch of Logarithm

If $G \subseteq \mathbb{C} \setminus \{0\}$ is a non-empty open set, then a *branch of log* is a continuous function $\ell : G \rightarrow \mathbb{C}$ such that for all $z \in G$, $\ell(z)$ is a logarithm of z , i.e. $e^{\ell(z)} = z$

Note

A branch of log may or may not exist depending on the shape of G .

Example 2.1

$G := \mathbb{C} \setminus (-\infty, 0]$. $\ell(z) = \ln|z| + i \operatorname{Arg}(z)$. Then

$$\begin{aligned} e^{\ell(z)} &= e^{\ln|z|}(\cos(\operatorname{Arg}(z)) + i \sin(\operatorname{Arg}(z))) \\ &= |z|(\cos(\operatorname{Arg}(z)) + i \sin(\operatorname{Arg}(z))) \\ &= z \end{aligned}$$

This is called the *principal branch of log* and is denoted by $\operatorname{Log}(z)$

Note

If we define $\ell(z) = \operatorname{Log}(z) + i2\pi k$ on $\mathbb{C} \setminus (-\infty, 0]$ for fixed $k \in \mathbb{Z}$, then we get a branch of log.

Remark 2.4

It is not possible to find a branch of log on $\mathbb{C} \setminus \{0\}$.

2.5 Further Holomorphicity Criteria**Exercise 2.5**

In polar form, the Cauchy-Riemann equations become, if $f(z) = u(z) + iv(z)$

$$\begin{aligned} ru_r &= v_\theta \\ \text{and } u_\theta &= -rv_r \end{aligned}$$

Theorem 2.3**Derivative of Logarithm**

If $\ell : G \rightarrow \mathbb{C}$ is a branch of log, then ℓ is holomorphic and $\ell'(z) = \frac{1}{z}$.

Theorem 2.4**Polynomial Holomorphicity Criterion**

Suppose f is a polynomial in variables x and y (with complex coefficients), then f is holomorphic iff f can be written as polynomial in $z = x + iy$.

3 Series**3.1 Numerical Series****Definition 3.1****Series**

A *series* is a formal sum $\sum_{n=0}^{\infty} c_n$ where the terms $c_n \in \mathbb{C}$. We say a series *converges* if the sequence of partial sums converge otherwise we say it *diverges*.

Lemma 3.1**Term Test**

If $\sum_{n=0}^{\infty} c_n$ converges, then $c_n \rightarrow 0$.

Definition 3.2**Absolute Convergence**

A series $\sum_{n=0}^{\infty} c_n$ converges absolutely if $\sum_{n=0}^{\infty} |c_n| < \infty$

Lemma 3.2**Absolute Convergence Test**

If $\sum_{n=0}^{\infty} c_n$ converges absolutely, then $\sum_{n=0}^{\infty} c_n$ converges and $|\sum_{n=0}^{\infty} c_n| \leq \sum_{n=0}^{\infty} |c_n|$

Example 3.1**Geometric Series**

$\sum_{n=0}^{\infty} c^n$ converges iff $|c| < 1$. If $|c| < 1$ then $\sum_{n=0}^{\infty} c^n = \frac{1}{1-c}$

3.2 Function Series**Definition 3.3****Convergence of Functions**

Suppose $(f_n)_{n=1}^{\infty}$ is a sequence of $\mathbb{C}$ -valued functions on $G \subseteq \mathbb{C}$.

- $(f_n)_{n=1}^{\infty}$ converges pointwise to f on $S \subseteq G$ if $\lim_{n \rightarrow \infty} f_n(z) = f(z)$ for every $z \in S$.
- $(f_n)_{n=1}^{\infty}$ converges uniformly to f on S if $\sup_{z \in S} |f_n(z) - f(z)| \rightarrow 0$
- $(f_n)_{n=1}^{\infty}$ converges locally uniformly to f on G if for each $z \in G$, there is some open ball $B \subseteq G$ with centre z such that $(f_n)_{n=1}^{\infty}$ converges uniformly to f on B .

Definition 3.4**Series of Functions**

If $(f_n)_{n=0}^{\infty}$ is a sequence of functions, $\sum_{n=0}^{\infty} f_n$ is the associated series of functions. The series converges to f if the partial sums do.

3.3 Power Series**Example 3.2**

$z_0 \in \mathbb{C}$ is fixed. $(f_n)_{n=0}^{\infty}$, where $f_n(z) = a_n(z - z_0)^n$ we get the series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$. The power series centred at z_0 .

Proposition 3.3**Region of Convergence**

The region of convergence for a power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ is of the form either

1. $\{z_0\}$
2. $\mathbb{C}$
3. The points in some disk centred at z_0 and possibly points on the boundary.

Example 3.3

$\sum_{n=0}^{\infty} z^n$ has radius of convergence 1. This represents the function $\frac{1}{1-z}$ on $\mathbb{D}$. This converges locally uniformly.

Theorem 3.4**Comparison Test for Power Series**

Suppose $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ and $\sum_{n=0}^{\infty} b_n(z - z_0)^n$ are power series with radius of convergence R_1, R_2 respectively. If $|b_n| \leq M|a_n|$ for some $M > 0$, then $R_1 \leq R_2$

Proof: This follows from the previous proposition because if $|z - z_0| < R$, where R is the radius of convergence, then $\sum a_n(z - z_0)^n$ converges absolutely. $\square$

Theorem 3.5**Cauchy-Hadamard Theorem**

The radius of convergence for a power series $\sum a_n(z - z_0)^n$ is

$$\frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}}$$

Theorem 3.6**Differentiation of Power Series**

Suppose $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ has radius of convergence $R > 0$. Let f be the function it represents (on that disk). Then the radius of convergence for $\sum_{n=1}^{\infty} n a_n(z - z_0)^{n-1}$ is also R and if g is the function it represents, then f is differentiable and $f'(z) = g(z)$ for $z \in G = \{z \in \mathbb{C} \mid |z - z_0| < R\}$

Proof: Note

$$\begin{aligned} & \frac{1}{\limsup_{n \rightarrow \infty} |n a_n|^{\frac{1}{n}}} \\ &= \frac{1}{\lim_{n \rightarrow \infty} n^{\frac{1}{n}}} \cdot \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}} \\ &= \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}} \\ &= R \end{aligned}$$

 $\square$ **Example 3.4**

$f(z) = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$ then f is entire, $f'(z) = f(z)$ and $f(0) = 1$.

3.4 Consequences of Power Series**Corollary 3.7****Smoothness of Power Series**

If f is represented by a power series on its disk of convergence, then derivatives of all orders exist on that disk.

4 Complex Integration

4.1 Complex Integrals on Intervals

Definition 4.1

Complex Integral

Suppose $\varphi : [a, b] \rightarrow \mathbb{C}$ is piecewise continuous (continuous except at finitely many points where one-sided limits exist). Then $\operatorname{Re} \varphi$ and $\operatorname{Im} \varphi$ are also piecewise continuous. Set

$$\int_a^b \varphi := \int_a^b \operatorname{Re} \varphi + i \int_a^b \operatorname{Im} \varphi$$

Note this is complex linear in φ

Theorem 4.1

Fundamental Theorem of Calculus

If φ is continuous and piecewise $\mathcal{C}^1$ on $[a, b]$, then

$$\int_a^b \varphi'(t) dt = \varphi(b) - \varphi(a)$$

Theorem 4.2

Integral Triangle Inequality

$$\left| \int_a^b \varphi(t) dt \right| \leq \int_a^b |\varphi(t)| dt$$

4.2 Path Integrals

Definition 4.2

Path

A path is a function $[a, b] \rightarrow \mathbb{C}$.

Definition 4.3

Piecewise $\mathcal{C}^1$ Path

A path $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise $\mathcal{C}^1$ if γ is continuous and there is a partition of $[a, b]$ such that γ is $\mathcal{C}^1$ on each subinterval, with one-sided limits for γ' at the breakpoints.

Definition 4.4

Path Integral

Suppose $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$ and $f : G \rightarrow \mathbb{C}$ is continuous on a set G containing the image of γ . Then define

$$\int_{\gamma} f(z) dz := \int_a^b f(\gamma(t)) \gamma'(t) dt$$

Example 4.1

$f(z) = ie^x$ where $z = x + iy$, $\gamma : [a, b] \rightarrow \mathbb{C}$, $\gamma(t) = t$ then

$$\int_{\gamma} f(z) dz = \int_a^b ie^t dt = i(e^b - e^a)$$

Theorem 4.3**Fundamental Theorem for Path Integrals**

If $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$ and if f is holomorphic on an open set that contains the range of γ then

$$\int_{\gamma} f'(z) dz = f(\gamma(b)) - f(\gamma(a))$$

Proof:

$$\int_{\gamma} f'(z) dz = \int_a^b f'(\gamma(t))\gamma'(t) dt = \int_a^b (f \circ \gamma)'(t) dt = (f \circ \gamma)(b) - (f \circ \gamma)(a)$$

□

Proposition 4.4**Reparametrization Invariance**

If $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$ and $\beta : [a_1, b_1] \rightarrow [a, b]$ is increasing and piecewise- $\mathcal{C}^1$ with $\beta(a_1) = a$ and $\beta(b_1) = b$. Set $\gamma_1 = \gamma \circ \beta$. Suppose f is continuous on the image of γ . Then

$$\int_{\gamma_1} f(z) dz = \int_{\gamma} f(z) dz.$$

Proof: By definition of the path integral and the chain rule,

$$\begin{aligned} \int_{\gamma_1} f(z) dz &= \int_{a_1}^{b_1} f(\gamma_1(s))\gamma_1'(s) ds \\ &= \int_{a_1}^{b_1} f(\gamma(\beta(s)))\gamma'(\beta(s))\beta'(s) ds \end{aligned}$$

By the substitution theorem with $t = \beta(s)$,

$$\begin{aligned} \int_{a_1}^{b_1} f(\gamma(\beta(s)))\gamma'(\beta(s))\beta'(s) ds &= \int_a^b f(\gamma(t))\gamma'(t) dt \\ &= \int_{\gamma} f(z) dz \end{aligned}$$

□

Proposition 4.5**Properties of Path Integrals**

1. If $a < c < b$ and $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$, let $\gamma_1 = \gamma|_{[a, c]}$ and $\gamma_2 = \gamma|_{[c, b]}$. Then

$$\begin{aligned} \int_{\gamma} f(z) \, dz &= \int_a^b f(\gamma(t)) \gamma'(t) \, dt \\ &= \int_a^c f(\gamma(t)) \gamma'(t) \, dt + \int_c^b f(\gamma(t)) \gamma'(t) \, dt \\ &= \int_{\gamma_1} f(z) \, dz + \int_{\gamma_2} f(z) \, dz \end{aligned}$$

2. If $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$, define the reverse path $-\gamma : [a, b] \rightarrow \mathbb{C}$ by $(-\gamma)(t) = \gamma(a + b - t)$. Then

$$\begin{aligned} \int_{-\gamma} f(z) \, dz &= \int_a^b f(\gamma(a + b - t)) \cdot (-\gamma'(a + b - t)) \, dt \\ &= - \int_a^b f(\gamma(a + b - t)) \gamma'(a + b - t) \, dt \\ &= - \int_a^b f(\gamma(s)) \gamma'(s) \, ds \\ &= - \int_{\gamma} f(z) \, dz \end{aligned}$$

Example 4.2

$z_0 \in \mathbb{C}$, $R > 0$, $n \in \mathbb{Z}$, $\gamma : [0, 2\pi] \rightarrow \mathbb{C}$, $\gamma(t) = z_0 + Re^{it}$. If $n \neq -1$, then

$$\int_{\gamma} (z - z_0)^n dz = 0$$

If $n = -1$, then

$$\int_{\gamma} (z - z_0)^n dz = 2\pi i$$

Indeed, if $n \neq -1$, then $F(z) = \frac{1}{n+1}(z - z_0)^{n+1}$ is an antiderivative of $(z - z_0)^n$, so

$$\begin{aligned} \int_{\gamma} (z - z_0)^n dz &= F(\gamma(2\pi)) - F(\gamma(0)) \\ &= \frac{1}{n+1}(\gamma(2\pi) - z_0)^{n+1} - \frac{1}{n+1}(\gamma(0) - z_0)^{n+1} \\ &= 0. \end{aligned}$$

If $n = -1$, then

$$\begin{aligned} \int_{\gamma} \frac{1}{z - z_0} dz &= \int_0^{2\pi} \frac{1}{\gamma(t) - z_0} \gamma'(t) dt \\ &= \int_0^{2\pi} \frac{1}{Re^{it}} iRe^{it} dt \\ &= \int_0^{2\pi} i dt \\ &= 2\pi i. \end{aligned}$$

4.3 Estimates and Uniform Convergence**Definition 4.5****Length of a Path**

If $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$ its length is

$$L(\gamma) := \int_a^b |\gamma'(t)| dt$$

Theorem 4.6**ML Inequality**

If f is continuous on the image of γ , then

$$\left| \int_{\gamma} f(z) dz \right| \leq L(\gamma) M_f$$

where M_f is the maximum value of $|(f \circ \gamma)(t)|$ for $t \in [a, b]$.

Proof:

$$\begin{aligned}
 \left| \int_{\gamma} f(z) \, dz \right| &= \left| \int_a^b f(\gamma(t)) \gamma'(t) \, dt \right| \\
 &\leq \int_a^b |f(\gamma(t)) \gamma'(t)| \, dt \\
 &= \int_a^b |f(\gamma(t))| |\gamma'(t)| \, dt \\
 &\leq \int_a^b M_f |\gamma'(t)| \, dt \\
 &= M_f L(\gamma)
 \end{aligned}$$

□

Theorem 4.7

Uniform Convergence for Path Integrals

Suppose $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$ and $(f_n)_{n=1}^{\infty}$ are continuous on the image of γ and (f_n) converges uniformly to f on the range of γ . Then

$$\int_{\gamma} f = \lim_{n \rightarrow \infty} \int_{\gamma} f_n$$

Proof: Since $f_n \rightarrow f$ uniformly and each f_n is continuous on the image of γ , f is continuous on the image of γ . By the ML inequality,

$$\begin{aligned}
 \left| \int_{\gamma} f_n - \int_{\gamma} f \right| &= \left| \int_{\gamma} (f_n - f) \right| \\
 &\leq L(\gamma) \sup_{t \in [a, b]} |f_n(\gamma(t)) - f(\gamma(t))| \\
 &= L(\gamma) \sup_{z \in \gamma([a, b])} |f_n(z) - f(z)|.
 \end{aligned}$$

Since $f_n \rightarrow f$ uniformly on the range of γ , the last expression tends to 0. Hence $\int_{\gamma} f_n \rightarrow \int_{\gamma} f$. □

4.4 Cauchy's Theorem

Example 4.3

Fourier Transform of the Gaussian

Let $a, b > 0$ and let $R_{a,b}$ be the positively oriented boundary of the rectangle with vertices $-a$, a , $a + ib$, and $-a + ib$. Consider

$$\int_{R_{a,b}} e^{-z^2} \, dz.$$

Since

$$e^{-z^2} = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{n!},$$

define

$$g(z) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)n!}.$$

The power series for g has infinite radius of convergence, and termwise differentiation gives $g'(z) = e^{-z^2}$. Hence

$$\int_{R_{a,b}} e^{-z^2} dz = \int_{R_{a,b}} g'(z) dz = 0.$$

Writing the four sides of $R_{a,b}$ separately gives

$$0 = \int_{-a}^a e^{-x^2} dx + \int_0^b e^{-(a+iy)^2} i dy - \int_{-a}^a e^{-(x+ib)^2} dx - \int_0^b e^{-(-a+iy)^2} i dy.$$

Thus

$$\int_{-a}^a e^{-(x+ib)^2} dx = \int_{-a}^a e^{-x^2} dx + i \int_0^b e^{-(a+iy)^2} dy - i \int_0^b e^{-(-a+iy)^2} dy.$$

Since

$$e^{-(x+ib)^2} = e^{-x^2+b^2-2ibx},$$

we get

$$\int_{-a}^a e^{-x^2} e^{-2ibx} dx = e^{-b^2} \int_{-a}^a e^{-x^2} dx + i e^{-b^2} \int_0^b e^{-(a+iy)^2} dy - i e^{-b^2} \int_0^b e^{-(-a+iy)^2} dy.$$

For $0 \leq y \leq b$,

$$|e^{-(a+iy)^2}| = e^{-a^2+y^2} \leq e^{-a^2+b^2}$$

and the same bound holds for $e^{-(-a+iy)^2}$. Hence the two vertical integrals tend to 0 as $a \rightarrow \infty$.

Therefore

$$\int_{-\infty}^{\infty} e^{-x^2} e^{-2ibx} dx = e^{-b^2} \int_{-\infty}^{\infty} e^{-x^2} dx.$$

Since $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$,

$$\int_{-\infty}^{\infty} e^{-x^2} e^{-2ibx} dx = \sqrt{\pi} e^{-b^2}.$$

Theorem 4.8**Cauchy's Theorem for a Triangle (Goursat's Lemma)**

If T is a triangle in $\mathbb{C}$ and if f is holomorphic in an open region that contains T and its interior, then

$$\int_T f(z) \, dz = 0.$$

Proof: Let

$$I = \int_T f(z) \, dz$$

and assume that T is positively oriented. Divide T into four congruent subtriangles by joining the midpoints of the sides. The boundary integrals over the four subtriangles add to I , because the integrals over the shared sides cancel. Hence one subtriangle, call it T_1 , satisfies

$$\left| \int_{T_1} f(z) \, dz \right| \geq \frac{|I|}{4}.$$

Repeating this construction gives nested triangles

$$T \supseteq T_1 \supseteq T_2 \supseteq \dots$$

such that

$$\left| \int_{T_n} f(z) \, dz \right| \geq \frac{|I|}{4^n}.$$

If P is the perimeter of T and D is the diameter of T , then the perimeter and diameter of T_n are $\frac{P}{2^n}$ and $\frac{D}{2^n}$ respectively.

Since the triangles are nested compact sets and their diameters tend to 0, there is a unique point

$$z_0 \in \bigcap_{n=1}^{\infty} T_n.$$

Since f is holomorphic at z_0 ,

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \eta(z)(z - z_0),$$

where $\eta(z) \rightarrow 0$ as $z \rightarrow z_0$. The first two terms have primitives, so their integrals around T_n are 0. Thus

$$\int_{T_n} f(z) \, dz = \int_{T_n} \eta(z)(z - z_0) \, dz.$$

Let $\varepsilon > 0$. Since $\eta(z) \rightarrow 0$, for all sufficiently large n we have $|\eta(z)| < \varepsilon$ for all $z \in T_n$. Hence, by the ML inequality,

$$\left| \int_{T_n} f(z) dz \right| \leq \varepsilon \cdot \left(\frac{D}{2^n} \right) \cdot \left(\frac{P}{2^n} \right) = \frac{\varepsilon PD}{4^n}.$$

Combining this with the lower bound gives, for all sufficiently large n ,

$$\frac{|I|}{4^n} \leq \left| \int_{T_n} f(z) dz \right| \leq \frac{\varepsilon PD}{4^n}.$$

Multiplying by 4^n , we get $|I| \leq \varepsilon PD$. Since $\varepsilon > 0$ was arbitrary, $I = 0$. □

4.5 Cauchy's Theorem for Convex Regions

Theorem 4.9

Cauchy's Theorem for Convex Regions

Suppose f is holomorphic in G and G is convex, then for any piecewise- $\mathcal{C}^1$ closed curve γ in G we have

$$\int_{\gamma} f(z) dz = 0$$

Proof: Fix $z_0 \in G$. For $z \in G$, define

$$F(z) = \int_{[z_0, z]} f(w) dw.$$

This is well-defined since G is convex. We claim that $F' = f$.

Let $z \in G$. Since G is open, for h sufficiently small we have $z + h \in G$. By convexity, the triangle with vertices $z_0, z, z + h$ is contained in G , so by Cauchy's theorem for triangles,

$$\int_{[z_0, z]} f(w) dw + \int_{[z, z+h]} f(w) dw - \int_{[z_0, z+h]} f(w) dw = 0.$$

Hence

$$F(z+h) - F(z) = \int_{[z, z+h]} f(w) dw.$$

Parametrizing the line segment $[z, z+h]$ by $w = z + th$, $0 \leq t \leq 1$, gives

$$\frac{F(z+h) - F(z)}{h} = \int_0^1 f(z+th) dt.$$

Therefore

$$\frac{F(z+h) - F(z)}{h} - f(z) = \int_0^1 (f(z+th) - f(z)) dt.$$

Since f is continuous at z , the right side tends to 0 as $h \rightarrow 0$. Thus $F'(z) = f(z)$.

Now if $\gamma : [a, b] \rightarrow G$ is a piecewise- $\mathcal{C}^1$ closed curve, then by [Theorem 4.3](#),

$$\int_{\gamma} f(z) dz = F(\gamma(b)) - F(\gamma(a)) = 0,$$

since $\gamma(a) = \gamma(b)$. □

4.6 Cauchy's Integral Formula

Theorem 4.10

Cauchy's Formula for the Circle

Suppose C is a circle oriented counter-clockwise and f is holomorphic in a region that contains C and its interior, then if z is in the interior of C

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta$$

Proof: Let

$$g(\zeta) = \begin{cases} \frac{f(\zeta) - f(z)}{\zeta - z} & \text{if } \zeta \neq z \\ f'(z) & \text{if } \zeta = z. \end{cases}$$

Then g is holomorphic in the interior of C , so by Cauchy's theorem for convex regions,

$$\int_C g(\zeta) d\zeta = 0.$$

Hence

$$\int_C \frac{f(\zeta)}{\zeta - z} d\zeta = f(z) \int_C \frac{1}{\zeta - z} d\zeta.$$

It remains to compute the last integral. Write $C(t) = z_0 + Re^{it}$, $0 \leq t \leq 2\pi$, and set $a = \frac{z - z_0}{R}$. Since z is in the interior of C , $|a| < 1$. Then

$$\begin{aligned} \int_C \frac{1}{\zeta - z} d\zeta &= \int_0^{2\pi} \frac{iRe^{it}}{z_0 + Re^{it} - z} dt \\ &= \int_0^{2\pi} \frac{i}{1 - ae^{-it}} dt \\ &= \int_0^{2\pi} i \sum_{n=0}^{\infty} a^n e^{-int} dt \\ &= 2\pi i. \end{aligned}$$

The third equality uses the geometric series expansion, valid uniformly since $|a| < 1$. The last equality follows by integrating the uniformly convergent series term-by-term: all terms with $n \geq 1$ integrate to 0, and the $n = 0$ term gives $2\pi i$. Therefore

$$\int_C \frac{f(\zeta)}{\zeta - z} d\zeta = 2\pi i f(z),$$

which gives the result. □

Note

The integral is 0 for z outside the circle

4.7 Consequences of Cauchy's Formula

Corollary 4.11

Mean Value Property

Suppose f is holomorphic on a region containing $\{z \in \mathbb{C} \mid |z - z_0| < R\}$. If $0 < r < R$, then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt.$$

Proof: Let C_r be the circle of radius r centered at z_0 , oriented counter-clockwise. By Cauchy's formula,

$$f(z_0) = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{\zeta - z_0} d\zeta.$$

Parametrize C_r by $\gamma(t) = z_0 + re^{it}$ for $0 \leq t \leq 2\pi$. Then $\gamma'(t) = ire^{it}$, so

$$\begin{aligned} f(z_0) &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + re^{it})}{re^{it}} ire^{it} dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt. \end{aligned}$$

□

Note

This is called the *mean value property* because $f(z_0)$ is equal to the average of the values of f on the circle centered at z_0 .

4.8 Cauchy Integrals and Analyticity

Definition 4.6

If φ is a continuous function on the range of some piecewise- $\mathcal{C}^1$ path γ . The *Cauchy Integral* over γ is the function

$$z \mapsto \int_{\gamma} \frac{\varphi(\zeta)}{\zeta - z} d\zeta$$

This is defined for z not in the range of γ

Proposition 4.12**Power Series of a Cauchy Integral**

Let γ be a piecewise- $\mathcal{C}^1$ path and suppose φ is continuous on the range of γ . Define

$$f(z) = \int_{\gamma} \frac{\varphi(\zeta)}{\zeta - z} d\zeta$$

for z not in the range of γ . If z_0 is not in the range of γ and

$$R = \text{dist}(z_0, \text{range}(\gamma)),$$

then f has a power series representation on $\{z \in \mathbb{C} \mid |z - z_0| < R\}$.

Proof: Let z satisfy $|z - z_0| < R$. Then for ζ in the range of γ ,

$$\left| \frac{z - z_0}{\zeta - z_0} \right| < 1.$$

Thus

$$\begin{aligned} \frac{1}{\zeta - z} &= \frac{1}{\zeta - z_0 - (z - z_0)} \\ &= \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \frac{z - z_0}{\zeta - z_0}} \\ &= \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}}. \end{aligned}$$

Therefore

$$f(z) = \int_{\gamma} \varphi(\zeta) \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}} d\zeta.$$

If $0 < \rho < R$ and $|z - z_0| \leq \rho$, then

$$\left| \frac{z - z_0}{\zeta - z_0} \right| \leq \frac{\rho}{R} < 1$$

for all ζ in the range of γ . Hence the geometric series converges uniformly on $|z - z_0| \leq \rho$ and on the range of γ , so we may integrate term by term. Thus

$$f(z) = \sum_{n=0}^{\infty} \left(\int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta \right) (z - z_0)^n.$$

This is a power series centered at z_0 . □

Theorem 4.13**Holomorphic Functions are Analytic**

Suppose f is holomorphic on a region containing $\{z \in \mathbb{C} \mid |z - z_0| < R\}$. Then f has a power series representation centered at z_0 with radius of convergence at least R . More precisely, for $0 < r < R$ and $|z - z_0| < r$,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n,$$

where

$$a_n = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta,$$

and C_r is the positively oriented circle of radius r centered at z_0 .

Proof: Fix $0 < r < R$. By Cauchy's formula, for $|z - z_0| < r$,

$$f(z) = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

This is a Cauchy integral with $\varphi(\zeta) = \frac{f(\zeta)}{2\pi i}$. By the power series representation for Cauchy integrals,

$$f(z) = \sum_{n=0}^{\infty} \left(\int_{C_r} \frac{f(\zeta)}{2\pi i (\zeta - z_0)^{n+1}} d\zeta \right) (z - z_0)^n.$$

Hence

$$a_n = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Since $r < R$ was arbitrary, the radius of convergence is at least R . □

4.9 Derivative Formula**Corollary 4.14****Cauchy's Integral Formula for Derivatives**

Suppose f is holomorphic on a region containing $\{z \in \mathbb{C} \mid |z - z_0| < R\}$. If $0 < r < R$ and C_r is the positively oriented circle of radius r centered at z_0 , then for every $n \geq 0$,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Proof: By the previous theorem, for $|z - z_0| < r$,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n,$$

where

$$a_n = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Since power series can be differentiated term-by-term inside their radius of convergence,

$$f^{(n)}(z_0) = n!a_n.$$

Substituting the formula for a_n gives

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

□

Corollary 4.15

Holomorphic Derivatives

If f is holomorphic on a region G , then $f^{(n)}$ is holomorphic on G for every $n \geq 0$.

Proof: Let $z_0 \in G$. Since G is open, there is $R > 0$ such that $\{z \in \mathbb{C} \mid |z - z_0| < R\} \subseteq G$. By the theorem above, f has a power series representation on this disk. Power series can be differentiated term-by-term inside their radius of convergence, so f' is holomorphic on the disk. Repeating this argument gives that $f^{(n)}$ is holomorphic near z_0 for every $n \geq 0$. Since $z_0 \in G$ was arbitrary, $f^{(n)}$ is holomorphic on G . □

4.10 Operations on Power Series

Proposition 4.16

Cauchy Product of Power Series

Suppose $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ has radius of convergence R_1 and $\sum_{n=0}^{\infty} b_n(z - z_0)^n$ has radius of convergence R_2 . Consider

$$\sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) (z - z_0)^n$$

Then the radius of convergence for this power series is at least $\min\{R_1, R_2\}$. If f and g are the functions defined by the first two power series, then for $|z - z_0| < \min\{R_1, R_2\}$,

$$f(z)g(z) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) (z - z_0)^n.$$

Proof: Let $R = \min\{R_1, R_2\}$ and define $h(z) = f(z)g(z)$ for $|z - z_0| < R$. Since f and g are holomorphic on this disk, so is h . Thus h has a power series representation

$$h(z) = \sum_{n=0}^{\infty} c_n(z - z_0)^n$$

on $|z - z_0| < R$, where $c_n = h^{(n)}(z_0)/n!$. By Leibniz's rule,

$$\begin{aligned}
h^{(n)}(z_0) &= \sum_{k=0}^n \binom{n}{k} f^{(k)}(z_0) g^{(n-k)}(z_0) \\
&= \sum_{k=0}^n \frac{n!}{k!(n-k)!} \cdot k! a_k \cdot (n-k)! b_{n-k} \\
&= n! \sum_{k=0}^n a_k b_{n-k}.
\end{aligned}$$

Hence

$$c_n = \sum_{k=0}^n a_k b_{n-k}.$$

Therefore the Cauchy product represents $h = fg$ on $|z - z_0| < R$, so its radius of convergence is at least R . □

4.11 Converse Results

Proposition 4.17

Converse of Goursat's Lemma

If $f : G \rightarrow \mathbb{C}$ is continuous and $\int_T f = 0$ for every triangle T whose edges and interior are contained in G , then f is holomorphic.

Proof: It is enough to show that f is holomorphic in a neighbourhood of each point. Fix $z_0 \in G$ and choose a disk D centered at z_0 with $D \subseteq G$.

For $z \in D$, define

$$g(z) = \int_{[z_0, z]} f(\zeta) d\zeta.$$

The triangle hypothesis implies that this integral is path independent in D : if $z_1, z_2, z_3 \in D$, then the integral around the triangle with these vertices is 0.

By the same line-segment argument as in the proof of Cauchy's theorem for convex regions, $g' = f$ on D . Therefore g is holomorphic on D . Since holomorphic derivatives are holomorphic, $f = g'$ is holomorphic on D . Since z_0 was arbitrary, f is holomorphic on G . □

4.12 Liouville's Theorem and Algebraic Consequences

Theorem 4.18

Liouville's Theorem

Suppose $f : \mathbb{C} \rightarrow \mathbb{C}$ is entire and bounded. Then f is constant.

Proof: Suppose $|f(z)| \leq M$ for all $z \in \mathbb{C}$. Fix $z_0 \in \mathbb{C}$ and $r > 0$. Let C_r be the circle centered at z_0 with radius r , oriented counter-clockwise. By Cauchy's integral formula for derivatives,

$$f'(z_0) = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^2} d\zeta.$$

By the ML inequality,

$$|f'(z_0)| \leq \frac{1}{2\pi} \cdot \left(\frac{M}{r^2} \right) \cdot 2\pi r = \frac{M}{r}.$$

Since this holds for every $r > 0$, letting $r \rightarrow \infty$ gives $f'(z_0) = 0$. Since z_0 was arbitrary, $f' = 0$ on $\mathbb{C}$, so f is constant. $\square$

Theorem 4.19

The Fundamental Theorem of Algebra

If $p(z) = \sum_{k=0}^n a_k z^k$ is a nonconstant polynomial with complex coefficients and $a_n \neq 0$, then there exist $z_1, \dots, z_n \in \mathbb{C}$ such that

$$p(z) = a_n \prod_{k=1}^n (z - z_k).$$

Proof: It is enough to show that p has a zero, since after finding a zero z_1 , we may factor $p(z) = (z - z_1)q(z)$ and repeat by induction on the degree.

Suppose, for contradiction, that p has no zero. Then $\frac{1}{p}$ is entire. We claim that $\frac{1}{p}$ is bounded. Since $a_n \neq 0$,

$$p(z) = z^n \left(a_n + \frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right).$$

For $|z|$ sufficiently large, the term in parentheses has absolute value at least $\frac{|a_n|}{2}$, so

$$|p(z)| \geq \frac{|a_n|}{2} \cdot |z|^n.$$

Hence $\left| \frac{1}{p(z)} \right| \rightarrow 0$ as $|z| \rightarrow \infty$, so $\frac{1}{p}$ is bounded outside a large disk. On the closed disk, $\frac{1}{p}$ is continuous, hence bounded. Thus $\frac{1}{p}$ is bounded on $\mathbb{C}$.

By Liouville's theorem, $\frac{1}{p}$ is constant, so p is constant, contradicting that p is nonconstant. Therefore p has a zero, and the factorization follows by induction. $\square$

5 Zeros and the Identity Theorem

5.1 Orders of Zeros

Definition 5.1

Zero of Order n

Suppose f is holomorphic on G and $f(z_0) = 0$. We say z_0 is a *zero of order n* if

$$0 = f(z_0) = f'(z_0) = \dots = f^{(n-1)}(z_0)$$

but $f^{(n)}(z_0) \neq 0$. A zero of order 1 is called a *simple zero*.

Proposition 5.1

Infinite-Order Zero

Suppose G is open, f is holomorphic on G , and f has a zero of infinite order at $z_0 \in G$. Then $f = 0$ on the connected component of G containing z_0 .

Proof: Let H be the connected component of G containing z_0 . Define

$$A = \{z \in H \mid f \text{ is identically zero on some disk centered at } z\}.$$

Since f has a zero of infinite order at z_0 , every coefficient in the power series for f centered at z_0 is 0. Thus f is identically zero on some disk centered at z_0 , so $z_0 \in A$ and A is non-empty.

The set A is open in H by definition. We show that A is closed in H . Suppose $z_k \in A$ and $z_k \rightarrow z \in H$. For each $m \geq 0$, since f is identically zero near each z_k , we have

$$f^{(m)}(z_k) = 0.$$

Since $f^{(m)}$ is continuous, $f^{(m)}(z) = 0$. Hence all derivatives of f vanish at z , so the power series for f centered at z is identically zero on some disk centered at z . Thus $z \in A$.

Therefore A is non-empty, open, and closed in H . Since H is connected, $A = H$, so $f = 0$ on H . $\square$

Proposition 5.2

Finite-Order Zeros are Isolated

Suppose f is holomorphic on an open set G and z_0 is a zero of finite order n . Then there is a disk centered at z_0 on which z_0 is the only zero of f .

Proof: Since z_0 is a zero of order n , the power series for f centered at z_0 has the form

$$f(z) = \sum_{k=n}^{\infty} a_k (z - z_0)^k$$

with $a_n \neq 0$. Thus

$$f(z) = (z - z_0)^n g(z),$$

where

$$g(z) = \sum_{k=0}^{\infty} a_{n+k} (z - z_0)^k.$$

Then g is holomorphic near z_0 and $g(z_0) = a_n \neq 0$. By continuity, $g(z) \neq 0$ on some disk centered at z_0 . On this disk, the only zero of $f(z) = (z - z_0)^n g(z)$ is z_0 . $\square$

5.2 Identity Theorem

Corollary 5.3

Identity Theorem

Suppose f and g are holomorphic on an open set G . If f and g agree on a subset $S \subseteq G$ that has a limit point $z_0 \in G$, then $f = g$ on the connected component of G containing z_0 .

Proof: Let $h = f - g$. Then h is holomorphic on G and $h = 0$ on S . Since z_0 is a limit point of S , there is a sequence (z_n) in S with $z_n \rightarrow z_0$ and $z_n \neq z_0$. Since h is continuous, $h(z_0) = 0$.

If h had a zero of finite order at z_0 , then z_0 would be an isolated zero of h , contradicting that $z_n \rightarrow z_0$ with $z_n \neq z_0$ and $h(z_n) = 0$. Hence h has a zero of infinite order at z_0 . By the infinite-order zero proposition, $h = 0$ on the connected component of G containing z_0 . Thus $f = g$ there. $\square$

6 Convergence and Maximum Principles

6.1 Identity Theorem Applications

Example 6.1

Suppose $f : \mathbb{D} \rightarrow \mathbb{C}$ is holomorphic and $f\left(\frac{1}{n}\right) = \frac{1}{n^2}$ for all $n \in \mathbb{N}$. What is $f'\left(\frac{i}{2}\right)$?

Proof: Let $g(z) = z^2$. Then $f\left(\frac{1}{n}\right) = g\left(\frac{1}{n}\right)$ for all $n \in \mathbb{N}$, and $\frac{1}{n} \rightarrow 0 \in \mathbb{D}$. By the identity theorem, $f = g$ on $\mathbb{D}$. Hence

$$f'\left(\frac{i}{2}\right) = g'\left(\frac{i}{2}\right) = 2 \cdot \frac{i}{2} = i.$$

□

6.2 Locally Uniform Limits

Proposition 6.1

Weierstrass Theorem

Suppose $G \subseteq \mathbb{C}$ is open and $f_k : G \rightarrow \mathbb{C}$ is holomorphic for each $k \in \mathbb{N}$. If $f_k \rightarrow f$ locally uniformly on G , then f is holomorphic. Moreover, for each $n \geq 0$, $f_k^{(n)} \rightarrow f^{(n)}$ locally uniformly on G .

Proof: Fix $z_0 \in G$ and choose $R > 0$ such that the closed disk $\overline{B(z_0, R)} \subseteq G$. Let C be the positively oriented circle centered at z_0 with radius R .

If $z \in B(z_0, R)$, then by Cauchy's formula,

$$f_k(z) = \frac{1}{2\pi i} \int_C f_k \frac{\zeta}{\zeta - z} d\zeta.$$

Since $f_k \rightarrow f$ uniformly on C , the uniform convergence theorem for path integrals gives

$$f(z) = \lim_{k \rightarrow \infty} f_k(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Thus f is a Cauchy integral on $B(z_0, R)$, so f is holomorphic there. Since z_0 was arbitrary, f is holomorphic on G .

If $0 < r < R$ and $z \in B(z_0, r)$, then by Cauchy's integral formula for derivatives,

$$f_k^{(n)}(z) - f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f_k(\zeta) - f(\zeta)}{(\zeta - z)^{n+1}} d\zeta.$$

Since $|\zeta - z| \geq R - r$ for $\zeta \in C$ and $z \in B(z_0, r)$, the ML inequality gives

$$\left| f_k^{(n)}(z) - f^{(n)}(z) \right| \leq n! \frac{R}{(R - r)^{n+1}} \sup_{\zeta \in C} |f_k(\zeta) - f(\zeta)|.$$

The supremum tends to 0 by local uniform convergence, so $f_k^{(n)} \rightarrow f^{(n)}$ uniformly on $B(z_0, r)$. Since z_0 was arbitrary, the convergence is locally uniform on G . □

6.3 Maximum Modulus Principle

Theorem 6.2

Maximum Modulus Principle

If $G \subseteq \mathbb{C}$ is a region and $f : G \rightarrow \mathbb{C}$ is holomorphic and nonconstant, then $|f|$ has no local maximum in G .

Proof: Suppose $|f|$ has a local maximum at $z_0 \in G$. Choose $R > 0$ such that $\{z \in \mathbb{C} \mid |z - z_0| < R\} \subseteq G$ and

$$|f(z)| \leq |f(z_0)|$$

for all $|z - z_0| < R$.

If $f(z_0) = 0$, then $|f(z)| = 0$ on this disk, so f is locally constant. By the identity theorem, f is constant on G , a contradiction. Thus $f(z_0) \neq 0$.

For $0 < r < R$, the mean value property gives

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt.$$

Taking absolute values,

$$|f(z_0)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})| dt \leq |f(z_0)|.$$

Hence equality holds throughout. Since the continuous function

$$t \mapsto |f(z_0)| - |f(z_0 + re^{it})|$$

is nonnegative and has integral 0, it is identically 0. Thus $|f(z_0 + re^{it})| = |f(z_0)|$ for all t .

Equality in the triangle inequality also forces all values $f(z_0 + re^{it})$ to have the same argument as $f(z_0)$. Therefore

$$f(z_0 + re^{it}) = f(z_0)$$

for all t . Since this holds for every $0 < r < R$, $f = f(z_0)$ on the disk centered at z_0 with radius R . By the identity theorem, f is constant on G , contradicting the hypothesis. Therefore $|f|$ has no local maximum in G . □

Corollary 6.3

If $f : G \rightarrow \mathbb{C}$ is holomorphic, nonconstant and $K \subseteq G$ compact and nonempty, then $|f|$ attains its maximum on ∂K .

6.4 Schwarz's Lemma

Theorem 6.4

Schwarz's Lemma

Suppose $f : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic and $f(0) = 0$. Then $|f(z)| \leq |z|$ for all $z \in \mathbb{D}$. Moreover, if $|f(z_0)| = |z_0|$ for some $z_0 \in \mathbb{D}$, $z_0 \neq 0$, then there exists $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ such that $f(z) = \lambda z$ for all $z \in \mathbb{D}$.

Proof: Define

$$g(z) = \begin{cases} \frac{f(z)}{z} & \text{if } z \neq 0 \\ f'(0) & \text{if } z = 0. \end{cases}$$

Since $f(0) = 0$, the singularity at 0 is removable, so g is holomorphic on $\mathbb{D}$.

Fix $0 < r < 1$. On the circle $|z| = r$,

$$|g(z)| = \frac{|f(z)|}{|z|} < \frac{1}{r}.$$

By the maximum modulus principle, for $|z| \leq r$ we have $|g(z)| \leq \frac{1}{r}$. Letting $r \rightarrow 1$ gives $|g(z)| \leq 1$ for all $z \in \mathbb{D}$. Hence, for $z \neq 0$,

$$|f(z)| = |z||g(z)| \leq |z|,$$

and the inequality is also true at $z = 0$.

Now suppose $|f(z_0)| = |z_0|$ for some $z_0 \neq 0$. Then $|g(z_0)| = 1$, so $|g|$ has a local maximum in $\mathbb{D}$. By the maximum modulus principle, g is constant. Thus $g(z) = \lambda$ for some $\lambda \in \mathbb{C}$ with $|\lambda| = 1$, and so $f(z) = \lambda z$ for all $z \in \mathbb{D}$. $\square$

6.5 Minimum Modulus Principle

Proposition 6.5

Minimum Modulus Principle

Suppose $G \subseteq \mathbb{C}$ is a region and $f : G \rightarrow \mathbb{C}$ is holomorphic and nonconstant. If $|f|$ has a local minimum at $z_0 \in G$, then $f(z_0) = 0$.

Proof: Suppose $|f|$ has a local minimum at z_0 and $f(z_0) \neq 0$. Then f is nonzero on some disk centered at z_0 , so $\frac{1}{f}$ is holomorphic on that disk. Since $|f|$ has a local minimum at z_0 , $\left|\frac{1}{f}\right|$ has a local maximum at z_0 . By the maximum modulus principle, $\frac{1}{f}$ is constant on a disk centered at z_0 , so f is constant on that disk. By the identity theorem, f is constant on G , a contradiction. Thus $f(z_0) = 0$. $\square$

Example 6.2

The conclusion is possible: if $f(z) = z$ and $K = \overline{\mathbb{D}}$, then $|f|$ attains its minimum at 0, and $f(0) = 0$.

7 Laurent Series

7.1 Examples

Example 7.1

For $|z| < 1$,

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n.$$

For $|z| > 1$,

$$\frac{1}{1-z} = -\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} = -\sum_{n=0}^{\infty} z^{-n-1} = -\sum_{n=1}^{\infty} z^{-n}.$$

Example 7.2

Consider

$$\frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1}.$$

The Laurent expansion depends on the annulus.

If $|z| < 1$, then

$$\begin{aligned} \frac{1}{z-2} &= -\frac{1}{2} \cdot \frac{1}{1-\frac{z}{2}} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}, \\ -\frac{1}{z-1} &= \frac{1}{1-z} = \sum_{n=0}^{\infty} z^n. \end{aligned}$$

Hence

$$\frac{1}{(z-1)(z-2)} = \sum_{n=0}^{\infty} \left(1 - \frac{1}{2^{n+1}}\right) z^n.$$

If $1 < |z| < 2$, then

$$\begin{aligned} \frac{1}{z-2} &= -\frac{1}{2} \cdot \frac{1}{1-\frac{z}{2}} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}, \\ -\frac{1}{z-1} &= -\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} = -\sum_{n=0}^{\infty} z^{-n-1}. \end{aligned}$$

Hence

$$\frac{1}{(z-1)(z-2)} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}} - \sum_{n=1}^{\infty} z^{-n}.$$

If $|z| > 2$, then

$$\frac{1}{z-2} = \frac{1}{z} \cdot \frac{1}{1-\frac{2}{z}} = \sum_{n=0}^{\infty} 2^n z^{-n-1},$$

$$-\frac{1}{z-1} = -\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} = -\sum_{n=0}^{\infty} z^{-n-1}.$$

Hence

$$\frac{1}{(z-1)(z-2)} = \sum_{n=0}^{\infty} (2^n - 1) z^{-n-1}.$$

7.2 Definition and Convergence

Definition 7.1

Laurent Series

A *Laurent Series* centered at z_0 is a series of the form

$$\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

where $a_n \in \mathbb{C}$. This converges at z if both

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n \quad \text{and} \quad \sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}$$

converge. Equivalently, the symmetric partial sums

$$\sum_{n=-N}^N a_n (z - z_0)^n$$

converge as $N \rightarrow \infty$. In that case,

$$\lim_{N \rightarrow \infty} \sum_{n=-N}^N a_n (z - z_0)^n$$

is defined to be the sum of the Laurent series.

Theorem 7.1**Convergence of Laurent Series**

Let

$$\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

be a Laurent series. Define

$$R_2 = \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}}$$

and

$$R_1 = \limsup_{n \rightarrow \infty} |a_{-n}|^{\frac{1}{n}}.$$

Then the positive part converges absolutely and locally uniformly on

$$|z - z_0| < R_2,$$

and the negative part converges absolutely and locally uniformly on

$$|z - z_0| > R_1.$$

Hence, if $R_1 < R_2$, the Laurent series converges absolutely and locally uniformly on the annulus

$$R_1 < |z - z_0| < R_2,$$

and the symmetric partial sums

$$\sum_{n=-N}^N a_n (z - z_0)^n$$

converge locally uniformly there. This annulus is called the *annulus of convergence* of the Laurent series.

Proof: By the Cauchy-Hadamard theorem, the positive part is an ordinary power series with radius of convergence R_2 , so it converges absolutely and locally uniformly on $|z - z_0| < R_2$.

For the negative part, the root test gives

$$\limsup_{n \rightarrow \infty} |a_{-n} (z - z_0)^{-n}|^{\frac{1}{n}} = \frac{R_1}{|z - z_0|}.$$

Thus the negative part converges absolutely when $\frac{R_1}{|z - z_0|} < 1$, i.e. when $|z - z_0| > R_1$. The same estimate gives locally uniform convergence on compact subsets of $|z - z_0| > R_1$.

On every compact subannulus

$$R_1 < r \leq |z - z_0| \leq R < R_2,$$

both the positive and negative parts converge uniformly. Hence, if $R_1 < R_2$, the Laurent series converges absolutely and the symmetric partial sums converge locally uniformly on $R_1 < |z - z_0| < R_2$. □

Theorem 7.2

Differentiating Laurent Series

Suppose the Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$$

converges on the annulus $R_1 < |z - z_0| < R_2$. Then f is holomorphic on this annulus, and

$$f'(z) = \sum_{n=-\infty}^{\infty} n a_n(z - z_0)^{n-1}.$$

Proof: By the convergence theorem, the symmetric partial sums converge locally uniformly on the annulus. Each symmetric partial sum is holomorphic on the annulus, so by Weierstrass's theorem the limit f is holomorphic.

Again by Weierstrass's theorem, the derivatives of the symmetric partial sums converge locally uniformly to f' . Since

$$\frac{d}{dz} \sum_{n=-N}^N a_n(z - z_0)^n = \sum_{n=-N}^N n a_n(z - z_0)^{n-1},$$

it follows that

$$f'(z) = \sum_{n=-\infty}^{\infty} n a_n(z - z_0)^{n-1}.$$

□

7.3 Annuli

Theorem 7.3

Cauchy's Theorem for an Annulus

Suppose $0 \leq R_1 < R_2 \leq \infty$ and f is holomorphic on the annulus

$$\{z \in \mathbb{C} \mid R_1 < |z - z_0| < R_2\}.$$

For $R_1 < r < R_2$, let C_r be the positively oriented circle centered at z_0 with radius r . Then $\int_{C_r} f(z) dz$ is independent of r . In other words, if $R_1 < r < s < R_2$, then

$$\int_{C_r} f(z) dz = \int_{C_s} f(z) dz.$$

Proof: Let

$$I(r) = \int_{C_r} f(z) dz.$$

Using the parametrization $z = z_0 + re^{it}$, $0 \leq t \leq 2\pi$, we have

$$I(r) = \int_0^{2\pi} f(z_0 + re^{it}) ire^{it} dt.$$

The integrand has continuous partial derivatives in r and t , so we may differentiate under the integral sign:

$$I'(r) = \int_0^{2\pi} \frac{\partial}{\partial r} (f(z_0 + re^{it}) ire^{it}) dt.$$

Now

$$\begin{aligned} \frac{\partial}{\partial r} (f(z_0 + re^{it}) ire^{it}) &= f'(z_0 + re^{it}) ire^{2it} + f(z_0 + re^{it}) ie^{it} \\ &= \frac{\partial}{\partial t} (f(z_0 + re^{it}) e^{it}). \end{aligned}$$

Therefore

$$\begin{aligned} I'(r) &= \int_0^{2\pi} \frac{\partial}{\partial t} (f(z_0 + re^{it}) e^{it}) dt \\ &= f(z_0 + re^{2\pi i}) e^{2\pi i} - f(z_0 + r) \\ &= f(z_0 + r) - f(z_0 + r) \\ &= 0. \end{aligned}$$

Hence I is constant on (R_1, R_2) , so $\int_{C_r} f(z) dz$ is independent of r . □

7.4 Cauchy's Formula on an Annulus

Theorem 7.4

Cauchy's Formula for an Annulus

Suppose $0 \leq R_1 < R_2 \leq \infty$ and f is holomorphic on the annulus

$$\{z \in \mathbb{C} \mid R_1 < |z - z_0| < R_2\}.$$

For $R_1 < r < R_2$, let C_r be the positively oriented circle centered at z_0 with radius r . If

$$R_1 < r_1 < |w - z_0| < r_2 < R_2,$$

then

$$f(w) = \frac{1}{2\pi i} \int_{C_{r_2}} \frac{f(z)}{z - w} dz - \frac{1}{2\pi i} \int_{C_{r_1}} \frac{f(z)}{z - w} dz.$$

Proof: Fix w as above and define

$$g(z) = \begin{cases} \frac{f(z)-f(w)}{z-w} & \text{if } z \neq w \\ f'(w) & \text{if } z = w. \end{cases}$$

Then g is holomorphic on the annulus. By Cauchy's theorem for an annulus,

$$\int_{C_{r_1}} g(z) dz = \int_{C_{r_2}} g(z) dz.$$

Hence

$$\int_{C_{r_2}} \frac{f(z)}{z-w} dz - \int_{C_{r_1}} \frac{f(z)}{z-w} dz = f(w) \int_{C_{r_2}} \frac{1}{z-w} dz - f(w) \int_{C_{r_1}} \frac{1}{z-w} dz.$$

The first integral on the right is $2\pi i$ by Cauchy's formula for the circle, since w lies inside C_{r_2} . The second integral is 0 by Cauchy's theorem for convex regions, since w lies outside C_{r_1} and $z \mapsto \frac{1}{z-w}$ is holomorphic on the disk bounded by C_{r_1} . Therefore

$$\int_{C_{r_2}} \frac{f(z)}{z-w} dz - \int_{C_{r_1}} \frac{f(z)}{z-w} dz = 2\pi i f(w),$$

which gives the formula. □

8 Isolated Singularities

8.1 Laurent Series Representation

Corollary 8.1

Laurent Series Representation

Suppose $0 \leq R_1 < R_2 \leq \infty$ and f is holomorphic on the annulus

$$A = \{z \in \mathbb{C} \mid R_1 < |z - z_0| < R_2\}.$$

Then f has a Laurent series representation on A :

$$f(w) = \sum_{n=-\infty}^{\infty} a_n (w - z_0)^n.$$

If $R_1 < r < R_2$ and C_r is the positively oriented circle centered at z_0 with radius r , then

$$a_n = \frac{1}{2\pi i} \int_{C_r} \frac{f(z)}{(z - z_0)^{n+1}} dz$$

for every $n \in \mathbb{Z}$.

Proof: Fix $w \in A$, and choose r_1, r_2 such that

$$R_1 < r_1 < |w - z_0| < r_2 < R_2.$$

By Cauchy's formula for an annulus,

$$f(w) = \frac{1}{2\pi i} \int_{C_{r_2}} \frac{f(z)}{z-w} dz - \frac{1}{2\pi i} \int_{C_{r_1}} \frac{f(z)}{z-w} dz.$$

On C_{r_2} we have $|w - z_0| < |z - z_0|$, so

$$\frac{1}{z-w} = \frac{1}{z-z_0} \cdot \frac{1}{1 - \frac{w-z_0}{z-z_0}} = \sum_{n=0}^{\infty} \frac{(w-z_0)^n}{(z-z_0)^{n+1}},$$

locally uniformly in w . Hence the outer integral contributes

$$\sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{C_{r_2}} \frac{f(z)}{(z-z_0)^{n+1}} dz \right) (w-z_0)^n.$$

On C_{r_1} we have $|z - z_0| < |w - z_0|$, so

$$\frac{1}{z-w} = -\frac{1}{w-z_0} \cdot \frac{1}{1 - \frac{z-z_0}{w-z_0}} = -\sum_{n=0}^{\infty} \frac{(z-z_0)^n}{(w-z_0)^{n+1}},$$

locally uniformly in w . Because the inner integral is subtracted, this contributes

$$\sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{C_{r_1}} f(z)(z-z_0)^n dz \right) (w-z_0)^{-n-1}.$$

These are exactly the Laurent coefficients above. The coefficient formula is independent of the chosen radius r by Cauchy's theorem for an annulus applied to $z \mapsto \frac{f(z)}{(z-z_0)^{n+1}}$. $\square$

8.2 Classification

Definition 8.1

Isolated Singularity

Suppose $f : G \rightarrow \mathbb{C}$ is holomorphic on a region G . A point z_0 is an *isolated singularity* of f if $z_0 \notin G$ and there is some $R > 0$ such that

$$\{z \in \mathbb{C} \mid 0 < |z - z_0| < R\} \subseteq G.$$

Note

If z_0 is an isolated singularity of f , then for sufficiently small $R > 0$, f is holomorphic on the punctured disk

$$0 < |z - z_0| < R.$$

Thus f has a Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

near z_0 .

Definition 8.2**Removable Singularity**

An isolated singularity z_0 of $f : G \rightarrow \mathbb{C}$ is *removable* if there is a holomorphic function $F : G \cup \{z_0\} \rightarrow \mathbb{C}$ such that $F|_G = f$.

Proposition 8.2**Removable Singularity Criterion**

Let z_0 be an isolated singularity of f , and write the Laurent series of f near z_0 as

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n.$$

Then z_0 is removable if and only if $a_n = 0$ for all $n < 0$.

Proof: If z_0 is removable, then the holomorphic extension has a power series at z_0 , so the Laurent series has no negative-power terms. Conversely, if all negative-power terms vanish, then the Laurent series is a power series in a disk centered at z_0 , so defining $F(z_0) = a_0$ gives a holomorphic extension. $\square$

Example 8.1

The function $f(z) = z$ on $\mathbb{C} \setminus \{0\}$ has a removable singularity at 0.

Example 8.2

The function

$$f(z) = \frac{z}{e^z - 1}$$

on $\mathbb{C} \setminus \{0\}$ has a removable singularity at 0, since $e^z - 1 = z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$

Definition 8.3**Pole**

An isolated singularity z_0 of f is a *pole of order m* if, in the Laurent series of f near z_0 ,

$$f(z) = \sum_{n=-m}^{\infty} a_n(z - z_0)^n$$

with $a_{-m} \neq 0$. A pole of order 1 is called a *simple pole*.

Proposition 8.3**Characterization of Poles**

Let z_0 be an isolated singularity of f . Then f has a pole of order m at z_0 if and only if $(z - z_0)^m f(z)$ has a removable singularity at z_0 and its removable extension is nonzero at z_0 .

Proof: If f has a pole of order m , then

$$f(z) = \sum_{n=-m}^{\infty} a_n(z - z_0)^n$$

with $a_{-m} \neq 0$. Hence

$$(z - z_0)^m f(z) = \sum_{n=-m}^{\infty} a_n (z - z_0)^{n+m} = \sum_{k=0}^{\infty} a_{k-m} (z - z_0)^k,$$

so $(z - z_0)^m f(z)$ has a removable singularity at z_0 , and the value of its extension at z_0 is $a_{-m} \neq 0$.

Conversely, if $(z - z_0)^m f(z)$ has a removable singularity at z_0 with nonzero value, write its holomorphic extension as

$$g(z) = \sum_{k=0}^{\infty} b_k (z - z_0)^k$$

with $b_0 \neq 0$. Then

$$f(z) = (z - z_0)^{-m} g(z) = \sum_{k=0}^{\infty} b_k (z - z_0)^{k-m},$$

so the first nonzero term is the $(z - z_0)^{-m}$ term. Thus f has a pole of order m at z_0 . $\square$

Example 8.3

The function $f(z) = \frac{1}{z}$ has a simple pole at 0.

Definition 8.4

Essential Singularity

An isolated singularity z_0 of f is an *essential singularity* if the Laurent series of f near z_0 has infinitely many nonzero negative-power terms.

Proposition 8.4

Classification of Isolated Singularities

Every isolated singularity is exactly one of the following: removable, a pole, or essential.

Proof: Let

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

be the Laurent series near the isolated singularity z_0 . If $a_n = 0$ for all $n < 0$, then z_0 is removable. If there are finitely many nonzero negative-power terms, let $-m$ be the smallest index with $a_{-m} \neq 0$. Then z_0 is a pole of order m . If there are infinitely many nonzero negative-power terms, then z_0 is essential. $\square$

Example 8.4

The function

$$f(z) = e^{\frac{1}{z}} = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n}$$

has an essential singularity at 0.

8.3 Singularity at Infinity**Definition 8.5****Isolated Singularity at Infinity**

Suppose $f : G \rightarrow \mathbb{C}$ is holomorphic. The point ∞ is an isolated singularity of f if there exists some $R > 0$ such that

$$\{z \in \mathbb{C} \mid |z| > R\} \subseteq G.$$

Definition 8.6**Classification at Infinity**

Suppose ∞ is an isolated singularity of f . Define

$$g(w) = f\left(\frac{1}{w}\right)$$

for $0 < |w| < \frac{1}{R}$. We say that ∞ is a removable singularity, a pole of order m , or an essential singularity of f if 0 is respectively a removable singularity, a pole of order m , or an essential singularity of g .

Example 8.5

The function $f(z) = \frac{1}{z}$ has a removable singularity at ∞ , since $f\left(\frac{1}{w}\right) = w$ has a removable singularity at 0.